MAT2002 Key to HW3

Spring 2026

Problem 1

Solve the IVP: $y'' + 2y' + 2y = 0$, $y(\pi/4) = 2$, $y'(\pi/4) = -2$

Solution: The characteristic equation is $r^2 + 2r + 2 = 0$, which gives roots $r = -1 \pm i$. The general solution to the homogeneous equation is:

$$y(t) = e^{-t}(c_1 \cos t + c_2 \sin t)$$

Taking the derivative:

$$y'(t) = -e^{-t}(c_1 \cos t + c_2 \sin t) + e^{-t}(-c_1 \sin t + c_2 \cos t)$$

Using the initial conditions:

$$y(\pi/4) = e^{-\pi/4}\left(c_1\frac{\sqrt{2}}{2} + c_2\frac{\sqrt{2}}{2}\right) = 2 \implies c_1 + c_2 = 2\sqrt{2}e^{\pi/4}$$

$$y'(\pi/4) = -y(\pi/4) + e^{-\pi/4}\left(-c_1\frac{\sqrt{2}}{2} + c_2\frac{\sqrt{2}}{2}\right) = -2$$

Substituting $y(\pi/4) = 2$:

$$-2 + e^{-\pi/4}\frac{\sqrt{2}}{2}(c_2 - c_1) = -2 \implies c_2 - c_1 = 0 \implies c_1 = c_2$$

Since $2c_1 = 2\sqrt{2}e^{\pi/4}$, we have $c_1 = c_2 = \sqrt{2}e^{\pi/4}$.

$$y(t) = \sqrt{2}e^{\pi/4}e^{-t}(\cos t + \sin t)$$

Problem 2

Solve the IVP: $y'' + y = 0$, $y(\pi/3) = 2$, $y'(\pi/3) = -4$

Solution: The general solution is $y(t) = c_1 \cos t + c_2 \sin t$.
Taking the derivative, $y'(t) = -c_1 \sin t + c_2 \cos t$.
Applying initial conditions:

$$y(\pi/3) = \frac{1}{2}c_1 + \frac{\sqrt{3}}{2}c_2 = 2 \implies c_1 + \sqrt{3}c_2 = 4$$
$$y'(\pi/3) = -\frac{\sqrt{3}}{2}c_1 + \frac{1}{2}c_2 = -4 \implies -\sqrt{3}c_1 + c_2 = -8$$

From the first equation, $c_1 = 4 - \sqrt{3}c_2$. Substitute this into the second:

$$-\sqrt{3}(4 - \sqrt{3}c_2) + c_2 = -8 \implies -4\sqrt{3} + 4c_2 = -8 \implies c_2 = \sqrt{3} - 2$$

Substitute c_2 back to find c_1:

$$c_1 = 4 - \sqrt{3}(\sqrt{3} - 2) = 1 + 2\sqrt{3}$$
$$y(t) = (1 + 2\sqrt{3}) \cos t + (\sqrt{3} - 2) \sin t$$

Problem 3

Solve the following for $t > 0$:
(i) $t^2y'' + 3ty' - 3y = 0$
(ii) $t^2y'' + 7ty' + 10y = 0$

Solution (i): This is a Cauchy-Euler equation. Let $y = t^r$.

$$r(r-1) + 3r - 3 = 0 \implies r^2 + 2r - 3 = 0 \implies (r+3)(r-1) = 0$$

Roots are $r = 1, -3$. General solution:

$$y(t) = c_1 t + c_2 t^{-3}$$

Solution (ii): Let $y = t^r$.

$$r(r-1) + 7r + 10 = 0 \implies r^2 + 6r + 10 = 0$$

Roots are $r = \frac{-6 \pm \sqrt{36-40}}{2} = -3 \pm i$. General solution:

$$y(t) = t^{-3}(c_1 \cos(\ln t) + c_2 \sin(\ln t))$$

Problem 4

Solve the following IVP:
(i) $9y'' + 6y' + 82y = 0$, $y(0) = -1$, $y'(0) = 2$
(ii) $y'' + 4y' + 4y = 0$, $y(-1) = 2$, $y'(-1) = 1$

Solution (i): Characteristic equation $9r^2 + 6r + 82 = 0$.

$$r = \frac{-6 \pm \sqrt{36 - 4(9)(82)}}{18} = \frac{-6 \pm \sqrt{-2916}}{18} = -\frac{1}{3} \pm 3i$$

General solution: $y(t) = e^{-t/3}(c_1 \cos 3t + c_2 \sin 3t)$. Using $y(0) = -1$, we get $c_1 = -1$.

$$\begin{aligned} y'(t) &= -\frac{1}{3}e^{-t/3}(-\cos 3t + c_2 \sin 3t) + e^{-t/3}(3 \sin 3t + 3c_2 \cos 3t) \\ y'(0) &= -\frac{1}{3}(-1) + 3c_2 = 2 \implies 3c_2 = \frac{5}{3} \implies c_2 = \frac{5}{9} \\ y(t) &= e^{-t/3}\left(-\cos 3t + \frac{5}{9}\sin 3t\right) \end{aligned}$$

Solution (ii): Characteristic equation $r^2 + 4r + 4 = 0 \implies (r+2)^2 = 0 \implies r = -2$.
General solution: $y(t) = (c_1 + c_2 t)e^{-2t}$.

$$\begin{aligned} y(-1) &= (c_1 - c_2)e^2 = 2 \implies c_1 - c_2 = 2e^{-2} \\ y'(t) &= c_2 e^{-2t} - 2(c_1 + c_2 t)e^{-2t} \\ y'(-1) &= c_2 e^2 - 2(c_1 - c_2)e^2 = 1 \implies -2c_1 + 3c_2 = e^{-2} \end{aligned}$$

Solving the system gives $c_2 = 5e^{-2}$ and $c_1 = 7e^{-2}$.

$$y(t) = (7e^{-2} + 5e^{-2}t)e^{-2t} = (7 + 5t)e^{-2(t+1)}$$

Problem 5

In each of following, use the method of reduction of order to find a second solution of the given differential equation:
(i) $xy'' - y' + 4x^3y = 0$**,** $x > 0$**;** $y_1(x) = \sin(x^2)$
(ii) $(x-1)y'' - xy' + y = 0$**,** $x > 1$**,** $y_1(x) = e^x$

Solution (i): Standard form is $y'' - \frac{1}{x}y' + 4x^2y = 0$, so $P(x) = -1/x$.

Reduction of order formula:

$$y_2(x) = y_1(x) \int \frac{e^{-\int P(x)dx}}{[y_1(x)]^2} dx$$
$$e^{-\int(-1/x)dx} = e^{\ln x} = x$$
$$y_2(x) = \sin(x^2) \int \frac{x}{\sin^2(x^2)} dx$$

Let $u = x^2, du = 2xdx$. The integral becomes $\frac{1}{2}\int \csc^2(u)du = -\frac{1}{2}\cot(x^2)$.

$$y_2(x) = \sin(x^2)\left(-\frac{1}{2}\frac{\cos(x^2)}{\sin(x^2)}\right) = -\frac{1}{2}\cos(x^2)$$

We can choose: $y_2(x) = \cos(x^2)$.

Solution (ii): Standard form has $P(x) = -\frac{x}{x-1} = -1 - \frac{1}{x-1}$.

$$e^{-\int P(x)dx} = e^{\int\left(1+\frac{1}{x-1}\right)dx} = e^{x+\ln(x-1)} = (x-1)e^x$$
$$y_2(x) = e^x \int \frac{(x-1)e^x}{e^{2x}} dx = e^x \int (x-1)e^{-x} dx$$

Using integration by parts:

$$\int (x-1)e^{-x}dx = -(x-1)e^{-x} - e^{-x} = -xe^{-x}$$
$$y_2(x) = e^x(-xe^{-x}) = -x$$

We can choose: $y_2(x) = x$.

Problem 6

Find the general solution for $y'' - y' - 2y = \cosh 2t$. Hint: $\cosh t = (e^t + e^{-t})/2$

Solution:
Homogeneous roots: $r^2 - r - 2 = 0 \implies r = 2, -1$. $y_c(t) = c_1 e^{2t} + c_2 e^{-t}$.
RHS is $\frac{1}{2}e^{2t} + \frac{1}{2}e^{-2t}$.

For $\frac{1}{2}e^{2t}$, since $r = 2$ is a root, guess $Y_1(t) = Ate^{2t}$.

$$Y_1'' - Y_1' - 2Y_1 = 3Ae^{2t} = \frac{1}{2}e^{2t} \implies A = \frac{1}{6}$$

For $\frac{1}{2}e^{-2t}$, guess $Y_2(t) = Be^{-2t}$.

$$Y_2'' - Y_2' - 2Y_2 = (4B - (-2B) - 2B)e^{-2t} = 4Be^{-2t} = \frac{1}{2}e^{-2t} \implies B = \frac{1}{8}$$

$$y(t) = c_1 e^{2t} + c_2 e^{-t} + \frac{1}{6}te^{2t} + \frac{1}{8}e^{-2t}$$

Problem 7

Solve the IVP $y'' - 2y' - 3y = 3te^{2t}$, $y(0) = 1$, $y'(0) = 0$

Solution:
Homogeneous roots: $r^2 - 2r - 3 = 0 \implies r = 3, -1$. $y_c(t) = c_1 e^{3t} + c_2 e^{-t}$.

Particular guess: $Y(t) = (At + B)e^{2t}$.

$$Y' = (2At + A + 2B)e^{2t}, \quad Y'' = (4At + 4A + 4B)e^{2t}$$
$$Y'' - 2Y' - 3Y = e^{2t}[-3At + 2A - 3B] = 3te^{2t}$$

Matching coefficients: $-3A = 3 \implies A = -1$. And $2A - 3B = 0 \implies -2 - 3B = 0 \implies B = -2/3$.

$$y(t) = c_1 e^{3t} + c_2 e^{-t} - \left(t + \frac{2}{3}\right)e^{2t}$$

Applying initial conditions gives the system $c_1 + c_2 = \frac{5}{3}$ and $3c_1 - c_2 = \frac{7}{3}$. Solving gives $c_1 = 1$, $c_2 = \frac{2}{3}$.

$$y(t) = e^{3t} + \frac{2}{3}e^{-t} - \left(t + \frac{2}{3}\right)e^{2t}$$

Problem 8

Consider the equation

$$y'' - 3y' - 4y = 2e^{-t} \tag{1}$$

$y_1(t) = e^{-t}$ and $y_2(t) = e^{4t}$ are solutions of the corresponding homogeneous equation. Adapting the method of reduction of order, seek a solution of the nonhomogeneous equation of the form $Y(t) = v(t)y_1(t) = v(t)e^{-t}$ where $v(t)$ is to be determined.

Solution (i): Substitute $Y(t)$, $Y'(t)$, and $Y''(t)$ into Eq. (1) and show that $v(t)$ must satisfy $v'' - 5v' = 2$.
Given $Y = ve^{-t}$, we find $Y' = (v' - v)e^{-t}$, and $Y'' = (v'' - 2v' + v)e^{-t}$.

$$\begin{aligned} Y'' - 3Y' - 4Y &= e^{-t}[v'' - 2v' + v - 3v' + 3v - 4v] \\ &= e^{-t}[v'' - 5v'] = 2e^{-t} \end{aligned}$$

Thus, $v'' - 5v' = 2$.

Solution (ii): Let $w(t) = v'(t)$ and show that $w(t)$ must satisfy $w' - 5w = 2$. Solve this equation for $w(t)$.
Substituting $w = v'$ gives $w' - 5w = 2$. Using integrating factor $\mu = e^{-5t}$:

$$(we^{-5t})' = 2e^{-5t} \implies we^{-5t} = -\frac{2}{5}e^{-5t} + C_1$$

$$w(t) = -\frac{2}{5} + C_1 e^{5t}$$

Solution (iii): Integrate $w(t)$ to find $v(t)$ and then show that $Y(t) = -\frac{2}{5}te^{-t} + \frac{1}{5}c_1e^{4t} + c_2e^{-t}$.

$$v(t) = \int \left(-\frac{2}{5} + C_1 e^{5t}\right) dt = -\frac{2}{5}t + \frac{C_1}{5}e^{5t} + C_2$$

$$Y(t) = v(t)e^{-t} = -\frac{2}{5}te^{-t} + \frac{C_1}{5}e^{4t} + C_2e^{-t}$$

Problem 9

Determine the general solution of $y'' + \lambda^2 y = \sum_{m=1}^{N} a_m \sin m\pi t$ **where** $\lambda > 0$ **and** $\lambda \neq m\pi$ **for** $m = 1, ..., N$**.**

Solution:

Homogeneous solution: $y_c(t) = c_1 \cos(\lambda t) + c_2 \sin(\lambda t)$.

By superposition, find a particular solution for each $a_m \sin(m\pi t)$. Since $\lambda \neq m\pi$, guess $Y_m(t) = A_m \cos(m\pi t) + B_m \sin(m\pi t)$. Substituting yields $A_m = 0$ and $B_m(\lambda^2 - m^2\pi^2) = a_m$, giving $B_m = \frac{a_m}{\lambda^2 - m^2\pi^2}$.

$$y(t) = c_1 \cos(\lambda t) + c_2 \sin(\lambda t) + \sum_{m=1}^{N} \frac{a_m}{\lambda^2 - m^2\pi^2} \sin(m\pi t)$$

Problem 10

In each of following, verify that the given functions y_1 and y_2 satisfy the corresponding homogeneous equation; then find a particular solution of the given nonhomogeneous equation.
(i) $t^2y'' - 2y = 3t^2 - 1$, $t > 0$; $y_1(t) = t^2$, $y_2(t) = t^{-1}$
(ii) $ty'' - (1+t)y' + y = t^2e^{2t}$, $t > 0$; $y_1(t) = 1 + t$, $y_2(t) = e^t$

Solution (i):
Verify y_1: $t^2(2) - 2(t^2) = 0$. Verify y_2: $t^2(2t^{-3}) - 2(t^{-1}) = 0$.
Standard form is $y'' - \frac{2}{t^2}y = 3 - t^{-2}$, so $g(t) = 3 - t^{-2}$.
Wronskian $W(t^2, t^{-1}) = t^2(-t^{-2}) - (2t)(t^{-1}) = -3$.

Variation of parameters:

$$u_1' = \frac{-t^{-1}(3 - t^{-2})}{-3} = t^{-1} - \frac{1}{3}t^{-3} \implies u_1 = \ln t + \frac{1}{6}t^{-2}$$
$$u_2' = \frac{t^2(3 - t^{-2})}{-3} = -t^2 + \frac{1}{3} \implies u_2 = -\frac{1}{3}t^3 + \frac{1}{3}t$$
$$Y_p = u_1y_1 + u_2y_2 = t^2 \ln t + \frac{1}{6} - \frac{1}{3}t^2 + \frac{1}{3} = t^2 \ln t + \frac{1}{2} - \frac{1}{3}t^2$$

(Note: The $-\frac{1}{3}t^2$ can be absorbed into the homogeneous solution). Particular solution:

$$Y_p(t) = t^2 \ln t + \frac{1}{2}$$

Solution (ii):
Verify y_1: $t(0) - (1+t)(1) + (1+t) = 0$. Verify y_2: $te^t - (1+t)e^t + e^t = 0$.
Standard form gives $g(t) = te^{2t}$. Wronskian $W(1+t, e^t) = (1+t)e^t - e^t = te^t$.

$$u_1' = \frac{-e^t(te^{2t})}{te^t} = -e^{2t} \implies u_1 = -\frac{1}{2}e^{2t}$$
$$u_2' = \frac{(1+t)(te^{2t})}{te^t} = (1+t)e^t \implies u_2 = \int (1+t)e^t dt = te^t$$
$$Y_p = u_1y_1 + u_2y_2 = -\frac{1}{2}e^{2t}(1+t) + (te^t)e^t$$
$$Y_p(t) = \frac{1}{2}(t-1)e^{2t}$$